Air University
Mid Term Examination: Fall 2024

Subject: Calculus and Analytical Geometry(MA-110)
Date: Nov. 04, 2024
Course Instructor: Dr. Waseem Khan
Student Roll Number:_____

Total Marks: 100
Time: 0900-1100 hrs
Classes: BEET-I-A, BCE-I-A

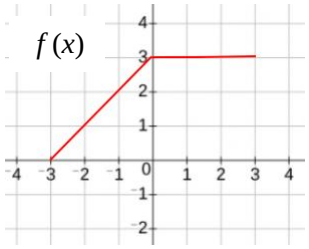
Special Instructions:

1. This is a closed book/closed notes examination. It contains ***03*** questions spanned over 02 page(s). Make sure that you have got the complete question paper.
2. **CALCULATORS ARE NOT ALLOWED.**
3. Use the given graph sheets for sketching the graphs in Q1 and Q2.
4. Attach the question paper with your answer book.

Question # 01: [CLO 01, PLO 1] **Marks: 10x5**

The graph of a piecewise defined function f(x) is sketched in Fig.1.

a. **Find** f (-1.5) from the graph.
b. **Express** the function mathematically and clearly specify the limits of the intervals.
c. Write down the domain and range of the function.
d. If g(x) = x^2, **evaluate** g(f (0)).
e. **Sketch** the function h(x) = f(x - 2) +1



Question # 02: [CLO 01, PLO 1] **Marks: 10x2**

Sketch the graphs of the given functions:

a. $g(x) = \cos\left(\frac{3\pi}{2} + x\right)$

b. $y = \sqrt{9 - x}$

Question # 03: [CLO 01, PLO 1] **Marks: 10x3**

Evaluate the limits of the given functions:

a. $\lim_{x \to -5} \frac{x^2 + 3x - 10}{x + 5}$

b. $\lim_{v \to 2} \frac{v^3 - 8}{v^4 - 16}$

c. $\lim_{x \to -1} \frac{\sqrt{x^2 + 8} - 3}{x + 1}$

